DELHI PUBLIC SCHOOL

Bokaro Steel City, Jharkhand

CLASS 6 BIOLOGY TERM-END EXAMINATION

SUBJECT: BIOLOGY	TIME ALLOWED: 1.5 HOURS	MAX MARKS: 35

Name: ______________________________ Roll No: _____________ Section: __________

General Instructions:

1. All questions are compulsory.
2. Read all questions carefully before answering.
3. Marks are indicated against each question.
4. For Section C, show all steps of your calculations including the formula, substitution, and final answer with appropriate units.

SECTION A — MULTIPLE CHOICE QUESTIONS

Select the most appropriate option for each of the following questions.

Q1. Which of the following raw materials is absorbed by plant roots from the soil to carry out the process of photosynthesis? **(1)**

(a) a) Carbon dioxide
(b) b) Oxygen
(c) c) Water
(d) d) Chlorophyll

Q2. Which adaptive feature helps a cactus plant survive and conserve water in a hot and dry desert habitat? **(1)**

(a) a) Broad and flat leaves to maximize sunlight absorption
(b) b) Modification of leaves into spines to reduce water loss via transpiration
(c) c) Soft and delicate green stems that easily bend with dry winds
(d) d) Shallow root system that dries up quickly after rain

Q3. A student performs an iodine test on a slice of raw potato. The potato slice turns blue-black in color. This change indicates the presence of which nutrient? **(1)**

(a) a) Proteins
(b) b) Fats
(c) c) Starch
(d) d) Vitamin C

Q4. Which type of joint is present in our shoulder, allowing bones to move freely in all directions? **(1)**

(a) a) Hinge joint
(b) b) Pivotal joint
(c) c) Ball and socket joint
(d) d) Fixed joint

SECTION B — SHORT QUESTIONS

Answer the following questions in 2 to 4 sentences.

Q1. Define the term 'Biotic Components' of an environment and give two examples. **(2)**

Q2. Explain why roots are considered a crucial part of a plant. State two main functions they perform. **(2)**

Q3. How does a frog adapt to live both on land and in water? **(2)**

SECTION C — NUMERICAL PROBLEMS

Solve the following quantitative biology problems. Show the formula used, step-by-step substitution, and the final answer with appropriate units.

Q1. During a physical education class, Rohan measures his pulse rate. He counts exactly 24 heartbeats in a span of 20 seconds. Calculate Rohan's heart rate in beats per minute (bpm). **(5)**

Q2. A student is observing an onion peel cell under a compound light microscope. The microscope has an eyepiece lens with a magnification of 15x and an objective lens with a magnification of 40x. Calculate the total magnification of the plant cell being observed. **(5)**

Q3. A Class 6 student planted a bean seed and measured its height daily. On Day 3, the shoot was 12 mm tall. On Day 10, the shoot grew to a height of 47 mm. Calculate the average growth rate of the bean shoot per day during this period. **(5)**

Q4. An active sixth-grader requires approximately 2000 kilocalories (kcal) of energy per day. If they obtain 60% of their daily energy requirement from carbohydrates, and 1 gram of carbohydrate provides 4 kcal of energy, calculate how many grams of carbohydrates the student needs to consume daily. **(5)**

Q5. A student observes a leaf epidermis under a microscope. The circular field of view has an area of 2 square millimeters (sq mm). If the student counts exactly 36 stomata within this field of view, calculate the stomatal density of the leaf surface in stomata per square millimeter. **(5)**

ANSWER KEY

SECTION A — MULTIPLE CHOICE QUESTIONS

Q1. c) Water [1 mark]

Q2. b) Modification of leaves into spines to reduce water loss via transpiration [1 mark]

Q3. c) Starch [1 mark]

Q4. c) Ball and socket joint [1 mark]

SECTION B — SHORT QUESTIONS

Q5. Biotic components refer to all the living organisms present in an ecosystem that interact with one another. Examples of biotic components include plants (autotrophs) and animals (consumers), which depend on each other and their abiotic environment for survival. [2 marks]

Q6. Roots are crucial because they perform vital functions necessary for a plant's survival. First, they firmly anchor the plant into the soil to provide structural support. Second, they absorb water and essential mineral nutrients from the soil, which are transported to the leaves for photosynthesis. [2 marks]

Q7. A frog has webbed feet that help it swim efficiently when in water, and strong hind limbs that allow it to leap on land. Furthermore, it can breathe through its moist skin while underwater, and uses its lungs to breathe air when it is on land. [2 marks]

SECTION C — NUMERICAL PROBLEMS

Q8. Formula: [5 marks]
Heart Rate (beats per minute) = (Number of Heartbeats / Time in seconds) * 60

Substitution:
Heart Rate = (24 / 20) * 60
Heart Rate = 1.2 * 60
Heart Rate = 72

Final Answer: 72 beats per minute (bpm)

Q9. Formula: [5 marks]
Total Magnification = Magnification of Eyepiece Lens * Magnification of Objective Lens

Substitution:
Total Magnification = 15 * 40
Total Magnification = 600

Final Answer: 600x (or 600 times magnification)

Q10. Formula: [5 marks]
Average Growth Rate = (Final Height - Initial Height) / (Final Day - Initial Day)

Substitution:
Average Growth Rate = (47 mm - 12 mm) / (10 - 3)
Average Growth Rate = 35 mm / 7 days
Average Growth Rate = 5 mm/day

Final Answer: 5 mm/day

Q11. Formula: [5 marks]
1. Energy from Carbohydrates (kcal) = Total Daily Energy Requirement * (Percentage / 100)
2. Mass of Carbohydrates (g) = Energy from Carbohydrates (kcal) / Energy provided per gram of carbohydrate

Substitution:
1. Energy from Carbohydrates = 2000 * (60 / 100) = 1200 kcal
2. Mass of Carbohydrates = 1200 kcal / 4 kcal/g = 300 g

Final Answer: 300 grams

Q12. Formula: [5 marks]
Stomatal Density = Total Number of Stomata / Area of the Field of View

Substitution:
Stomatal Density = 36 stomata / 2 sq mm
Stomatal Density = 18 stomata per sq mm

Final Answer: 18 stomata/sq mm